

**D.Y. PATIL COLLEGE OF ENGINEERING &  
TECHNOLOGY, KOLHAPUR**

**(An Autonomous Institute)**



**DEPARTMENT OF CSE (DATA SCIENCE)**

A

Project Report

On

**“MindTrack: Doctor–Patient Mental Health Task  
Assignment & Prediction”**

Submitted by

| <b>Name</b>                 | <b>Roll No.</b> |
|-----------------------------|-----------------|
| Miss. Saloni Rajan Naik     | 21              |
| Mr. Shivam Dnyanoba Katkade | 22              |
| Mr. Om Vijay Mahajan        | 23              |
| Mr. Sakib Husen Pendhari    | 46              |

**Under the guidance of**

**Ms. A. M. Chavan**

**Final Year B. Tech. CSE (Data Science)**

**Academic Year 2025-26**

# D. Y. PATIL COLLEGE OF ENGINEERING & TECHNOLOGY, KOLHAPUR

(An Autonomous Institute)



## DEPARTMENT OF CSE (DATA SCIENCE) CERTIFICATE

This is to certify that,

| Roll No. | Unique ID  | Student Name                | Exam Seat No. |
|----------|------------|-----------------------------|---------------|
| 21       | EN21129121 | Miss. Saloni Rajan Naik     | 25326         |
| 22       | EN22158098 | Mr. Shivam Dnyanoba Katkade | 25347         |
| 23       | EN22229362 | Mr. Om Vijay Mahajan        | 25288         |
| 46       | EN22177597 | Mr. Sakib Husen Pendhari    | 25323         |

Have successfully completed the project work entitled,

### “MindTrack: Doctor–Patient Mental Health Task Assignment & Prediction”

In partial fulfilment for the curriculum of **Final Year B. Tech. CSE (Data Science)**. This is the record of their work carried out during academic year 2024-2025.

**Date:**

**Place:** Kolhapur

**Guide Name**  
Project Guide

**Prof. DR. G. V. Patil**  
HOD

**Prof. DR. S. D. Chede**  
Principal

**External Examiner**

# DECLARATION

We the undersigned students of **Final Year B. Tech. CSE (Data Science)** declare that the project work report entitled “**MindTrack: Doctor–Patient Mental Health Task Assignment & Prediction**” written and submitted by us, under the guidance of **Ms. A. M. Chavan** is our original work. The empirical findings in this report are based on the data collected by us. The matter assimilated in this report is not the reproduction of any readymade report. We have not violated any of the provisions under the Copyright and Piracy / Cyber / IPR Act amended from time to time.

**Date:**

**Place:** Kolhapur

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This project report stands as a testament to the collective efforts and support extended by each individual mentioned above. We acknowledge and appreciate your contributions, which have enriched our learning experience and propelled us towards excellence.

**Date:**

**Place:** Kolhapur

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## ABSTRACT

MindTrack is a comprehensive Android application designed to enhance the communication and therapeutic process between psychologists and their patients. The application serves as a dedicated platform for patient management, task assignment, and psychological trend analysis. The core of the system lies in its dual functionality: a user- friendly interface for psychologists to assign homework or tasks to patients, and a powerful machine learning component that employs unsupervised learning to analyze patient data. By identifying common stressors, emotional patterns, or behavioral trends across an entire caseload, the system provides psychologists with invaluable insights. This trend analysis enables doctors to better understand recurring issues and to refine their therapeutic focus, ultimately leading to more effective and personalized treatment plans. The application aims to create a more efficient, data-driven, and proactive approach to mental healthcare.

# INTRODUCTION

## **a. Need of the Work**

Mental health issues such as stress, anxiety, depression, and emotional instability are increasing rapidly in modern society. Psychologists and therapists often face challenges in continuously monitoring a patient's emotional condition outside scheduled therapy sessions. Traditional therapy methods mainly depend on periodic face-to-face interactions, which provide limited information about a patient's day-to-day mental state and behavioral changes. As a result, therapists may not fully understand the effectiveness of treatments or identify emerging psychological concerns in time.

There is a growing need for a smart digital platform that enables continuous interaction between psychologists and patients while supporting therapeutic activities beyond clinical sessions. Additionally, mental health professionals require analytical tools that can help them identify common behavioral patterns, treatment effectiveness, and shared psychological challenges among patients. Therefore, a secure, intelligent, and data-driven mental healthcare system is essential to improve therapy outcomes and make mental health support more accessible and effective.

## **b. Hardware Requirements**

Existing mental health systems and therapy practices mainly focus on direct communication through appointments, video consultations, or simple task management features. Some applications provide mood tracking, journaling, reminders, or meditation exercises, but these systems generally work independently without offering deeper analytical insights. Most current platforms do not utilize machine learning techniques to study collective behavioral trends or therapy effectiveness.

Traditional systems also lack continuous patient monitoring and often fail to provide psychologists with a complete understanding of patient progress between therapy sessions. In many cases, data collected from users remains underutilized, and therapists must rely primarily on manual observations and patient feedback. Furthermore, several existing applications have limitations related to personalization, intelligent recommendations, and advanced privacy-focused data analysis.

### c. **Proposed System**

The proposed system, **MindTrack**, is an intelligent mental healthcare platform designed to strengthen the connection between psychologists and patients through continuous digital interaction. The system allows psychologists to assign therapeutic tasks such as journaling, mood tracking, mindfulness activities, and cognitive behavioral therapy exercises. Patients can regularly update their emotional status and complete assigned activities through the platform, enabling therapists to monitor progress more effectively.

A key feature of the proposed system is the integration of machine learning techniques, particularly unsupervised learning algorithms, to analyze anonymized patient data such as mood patterns, task completion behavior, and journal activity. Instead of diagnosing individuals, the system identifies broader trends, common stress factors, and therapy effectiveness across multiple patients. These insights help psychologists improve treatment strategies and focus on areas requiring additional attention or research.

The system also emphasizes strong data privacy and security through encryption and anonymization methods, ensuring that sensitive patient information remains protected. Additionally, the adaptive learning capability of the platform enables continuous improvement in analytical accuracy as more data becomes available. By combining communication, therapy management, secure data handling, and machine learning-based analytics, “MindTrack” aims to transform traditional reactive therapy into a proactive, personalized, and data-driven mental healthcare solution.



# LITERATURE SURVEY

The foundation for understanding mental health conditions is provided by the Diagnostic and Statistical Manual of Mental Disorders (5th ed., text rev.) published by the American Psychiatric Association [1]. This manual offers standardized diagnostic criteria and classification of mental disorders, serving as the clinical basis for designing applications that aim to monitor and support mental well-being. While it does not present a technological system, it provides the essential framework for identifying relevant symptoms, data points, and indicators that digital platforms can track.

Smith, Jones, and Lee [2] explored the use of sentiment analysis on patient journal entries for mental health monitoring in the Journal of Medical Internet Research. Their study highlights a system that processes textual data to extract emotional tone and sentiment over time, providing clinicians with insights into patient mood fluctuations and emotional health. This approach demonstrates the potential of natural language processing to transform unstructured patient input into meaningful trends that can be used for timely interventions.

Jones et al. [3], in their systematic review published in Digital Health, assessed the efficacy of existing mobile applications for mental health. Their review identified a wide range of features such as mood tracking, task reminders, journaling, and CBT-based interventions, but also noted significant limitations related to security, user engagement, and personalization. This review provides valuable guidance for future app design by highlighting both the strengths and shortcomings of current mobile health solutions.

Kim and Chen [4], writing in Psychological Science, applied unsupervised learning techniques for patient segmentation in digital mental health systems. Their study demonstrates how clustering algorithms can identify subgroups of patients with similar emotional and behavioural patterns. Such segmentation enables a more personalized approach to care, revealing population-level trends and stressors that can guide clinical decision-making. This work directly supports the integration of unsupervised models into mental health applications.

Finally, Sharma et al. [5] introduced the application of large language models for automated clinical documentation summarization in Nature Medicine. Their research describes systems capable of condensing lengthy patient records into concise, clinically relevant summaries. This innovation reduces the cognitive burden on psychologists and enhances efficiency by highlighting key symptoms, changes, and progress markers. Incorporating such summarization tools into digital platforms can greatly improve the quality of insights available to practitioners.

Disorders (DSM-5), which provides standardized criteria essential for designing digital mental health tools. Research by Smith, Jones, and Lee demonstrates the use of sentiment analysis on patient journals to monitor mood changes, showcasing the value of natural language processing in clinical settings. A systematic review by Jones et al. highlights common features and limitations of current mental health apps, emphasizing the need for improved security, engagement, and personalization. Kim and Chen's work on unsupervised learning shows how clustering techniques can segment patients based on behavioral patterns, enabling more tailored care. Lastly, Sharma et al. introduce large language models for automated summarization of clinical records, which can reduce clinician workload and enhance insight quality. Together, these studies support the integration of advanced machine learning and natural language processing into mental health platforms to improve monitoring, personalization, and clinical efficiency.

The literature survey highlights the growing importance of digital technologies in mental healthcare systems. Existing studies demonstrate that sentiment analysis, natural language processing, unsupervised machine learning, and large language models can significantly improve mental health monitoring, patient engagement, and clinical decision-making. Research on existing mental health applications also reveals limitations related to personalization, security, and analytical capabilities. These studies collectively support the development of intelligent platforms like MindTrack, which integrates mood tracking, task management, data analytics, and machine learning to provide a more proactive, secure, and data-driven approach to mental healthcare.

# PROBLEM STATEMENT & OBJECTIVES

## **a. problem statement:**

“To design & develop an MindTrack, android application with ML trend discovery.”

## **b. Objectives:**

1. To design and implement a user-friendly Android application with separate dashboards for psychologists and patients.
2. To enable secure task assignment from the psychologist to the patient, with real-time tracking of task completion.
3. To develop a data collection module that securely and anonymously logs patient submitted data (e.g., mood ratings, journal entries).
4. To implement an unsupervised machine learning model that analyzes anonymized data to identify and visualize trends in emotional patterns or stressors across a patient population.
5. To provide a clear, insightful report for the psychologist based on the machine learning analysis, guiding them on areas of focus for their practice.

# REQUIREMENT ANALYSIS

## a. Dataset Description

For the development and training of the analytical modules in MindTrack, publicly available mental health and emotion-analysis datasets were utilized from platforms such as [Kaggle](#) and other open research repositories. These datasets contain anonymized records related to emotional states, mood patterns, stress levels, mental health indicators, and user-generated textual inputs such as journal entries and daily reflections.

The dataset used for training primarily focuses on mood and emotional classification, where patient responses are categorized into emotional states such as *happy*, *sad*, *anxious*, *stressed*, *calm*, and *neutral*. The collected data includes attributes such as daily mood entries, sleep patterns, stress levels, activity records, task completion rates, and sentiment scores extracted from text-based journals. These features help the machine learning model identify behavioural and emotional trends over a weekly period.

To support predictive analysis and visualization, historical weekly data generated by users is processed and analyzed using machine learning and data analytics techniques. The system studies changes in mood patterns over time and generates insights regarding emotional stability, stress fluctuations, consistency in therapeutic activities, and overall mental well-being.

The datasets selected from Kaggle and related research sources were preprocessed using Python libraries such as Pandas, NumPy, and Scikit-learn. Data preprocessing steps included handling missing values, data normalization, categorical encoding, and feature extraction. The processed data was then used to train machine learning models capable of identifying patterns and generating weekly analytical summaries for psychologists and patients.

## b. Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB free space
- System Type: 64-bit Operating System
- Display: 1366 × 768 resolution or higher
- Android Device: For application testing
- Internet Connection: Required for Firebase and data synchronization

## c. Software Requirements

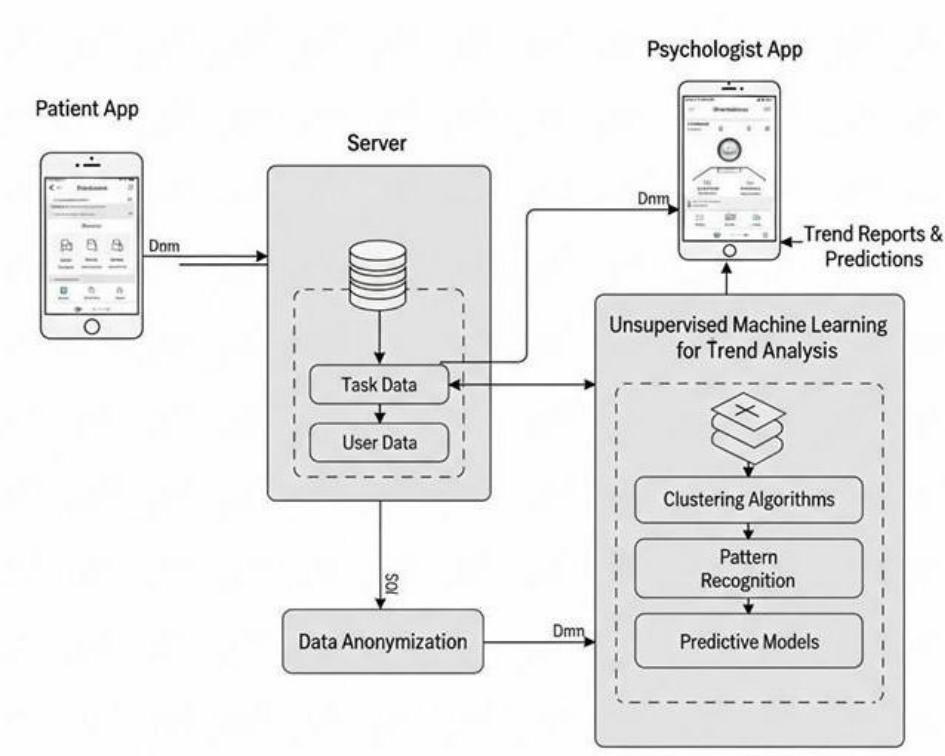
- Operating System: Windows 10/11
- Development Tool: Android Studio
- Programming Languages: Java, Python
- Database: Firebase
- Version Control: Git
- SDK: Android SDK
- Internet Browser: Google Chrome or any modern browser

## d. Libraries Used

- **Android SDK:** Core tools and APIs for building the MindTrack mobile app's user interface and functionality on Android devices.
- **Navigation Component:** To manage seamless navigation and screen transitions within the app for a smooth user experience.
- **Firebase SDK (Firestore, Authentication):** To securely store patient data in the cloud and manage user sign-in and authentication.
- **scikit-learn:** To perform unsupervised machine learning tasks like clustering patient data to identify patterns and trends.
- **Matplotlib/Seaborn:** To generate visual data representations (graphs, charts) that help psychologists interpret aggregated patient insights.

# SYSTEM DESIGN & IMPLEMENTATION

## a. System Architecture



**1. Patient App:** This is the mobile application used by patients. Its primary function is to allow patients to receive and view tasks or homework assigned by their psychologist, submit data (like mood logs or journal entries), and communicate securely.

**2. Psychologist App:** This is the interface for the psychologist. It allows them to create and assign tasks, monitor patient progress, and, most importantly, view the trend reports and predictions generated by the machine learning module.

**3. Server:** The central hub of the application. It handles all communication between the patient and psychologist apps, ensuring secure data exchange. It also manages the application's logic and serves as the bridge to the database and machine learning module.

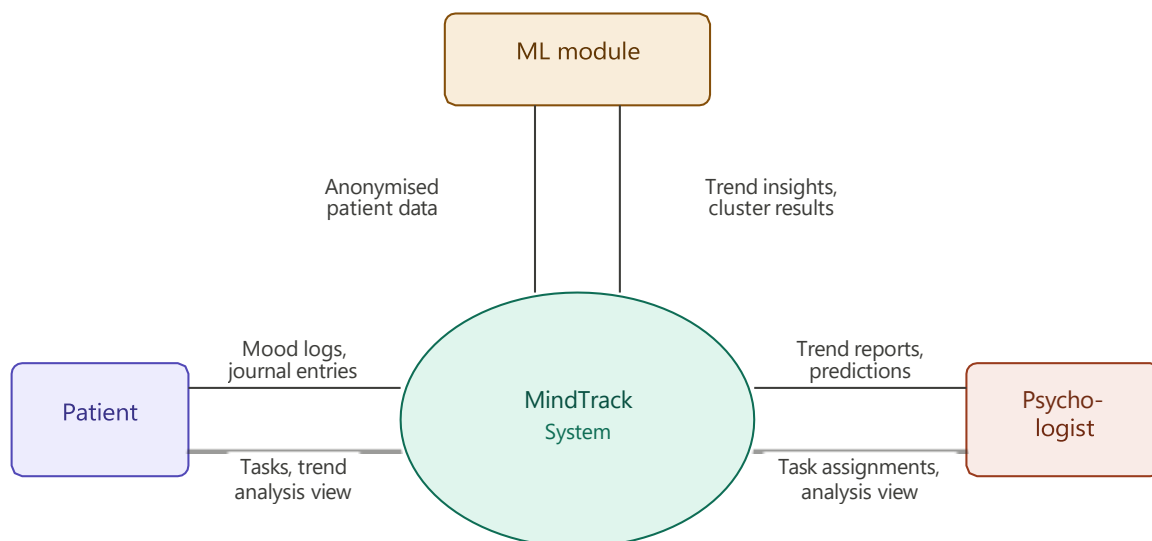
**4. Database (Task Data & User Data):** This stores all the critical information for the application. Task Data includes details about the homework assigned by psychologists, while User Data contains secure, anonymized information about patient progress, mood, and other relevant metrics.

**5. Data Anonymization:** This is a crucial step for patient privacy. It processes the raw data from the database, removing any personally identifiable information so that the data can be safely used for analysis by the machine learning module without compromising confidentiality.

**6. Unsupervised Machine Learning for Trend Analysis:** This is the core of the prediction functionality. It takes the anonymized data and uses algorithms like clustering and pattern recognition to discover hidden patterns and trends across the entire patient caseload. It's not about an individual patient but about the collective behaviors and emotional patterns.

**7. Trend Reports & Predictions:** This is the final output of the machine learning module. It's a summary of the insights gained from the data analysis. This report is sent back to the Psychologist App, helping the psychologist to identify common stressors or emotional patterns and better understand which areas of their practice need more focus.

## b. Data Flow Diagrams



Level 0 — Context diagram

**Fig. 6.1: DFD- Level 0**

MindTrack — DFD level 1

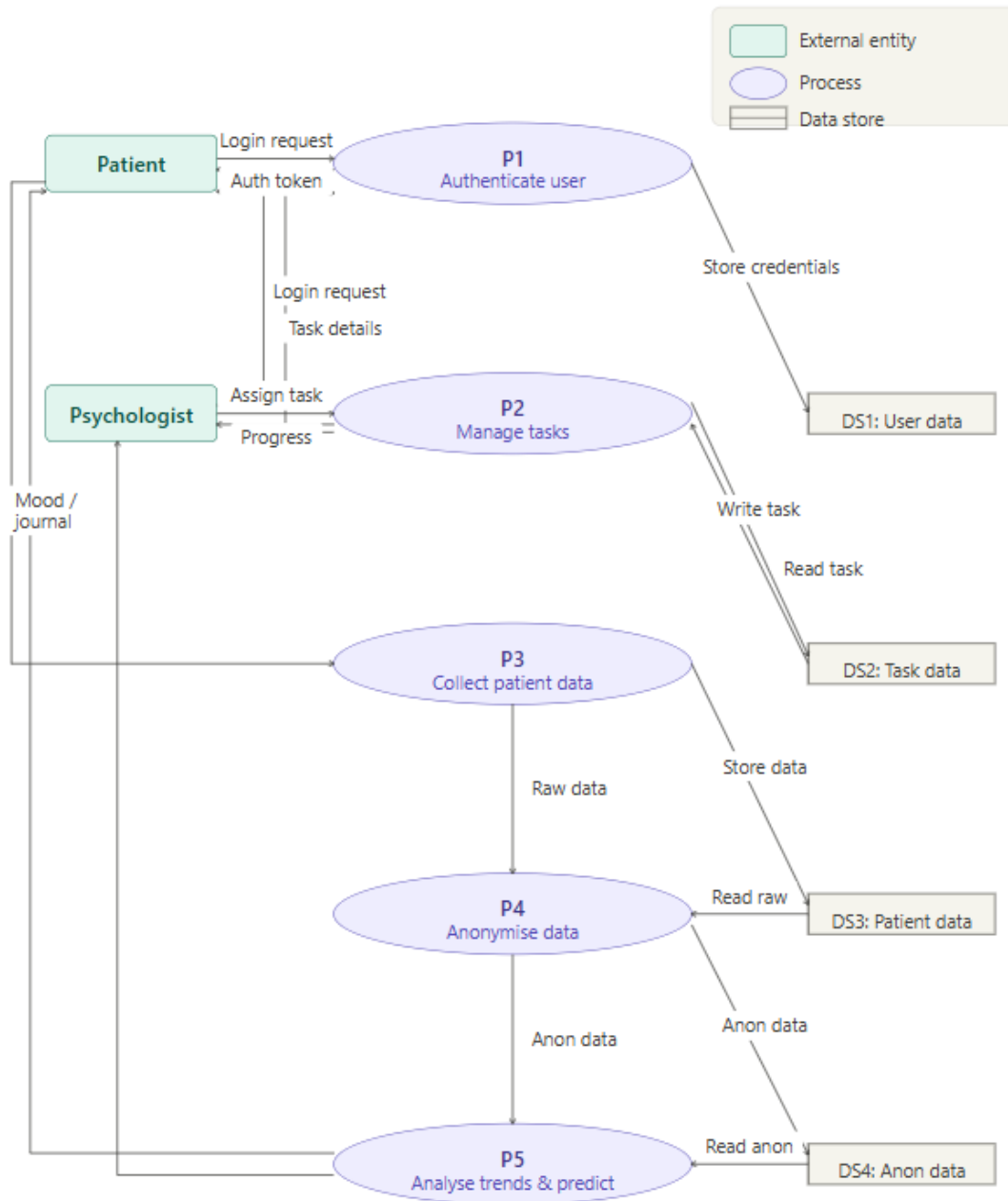
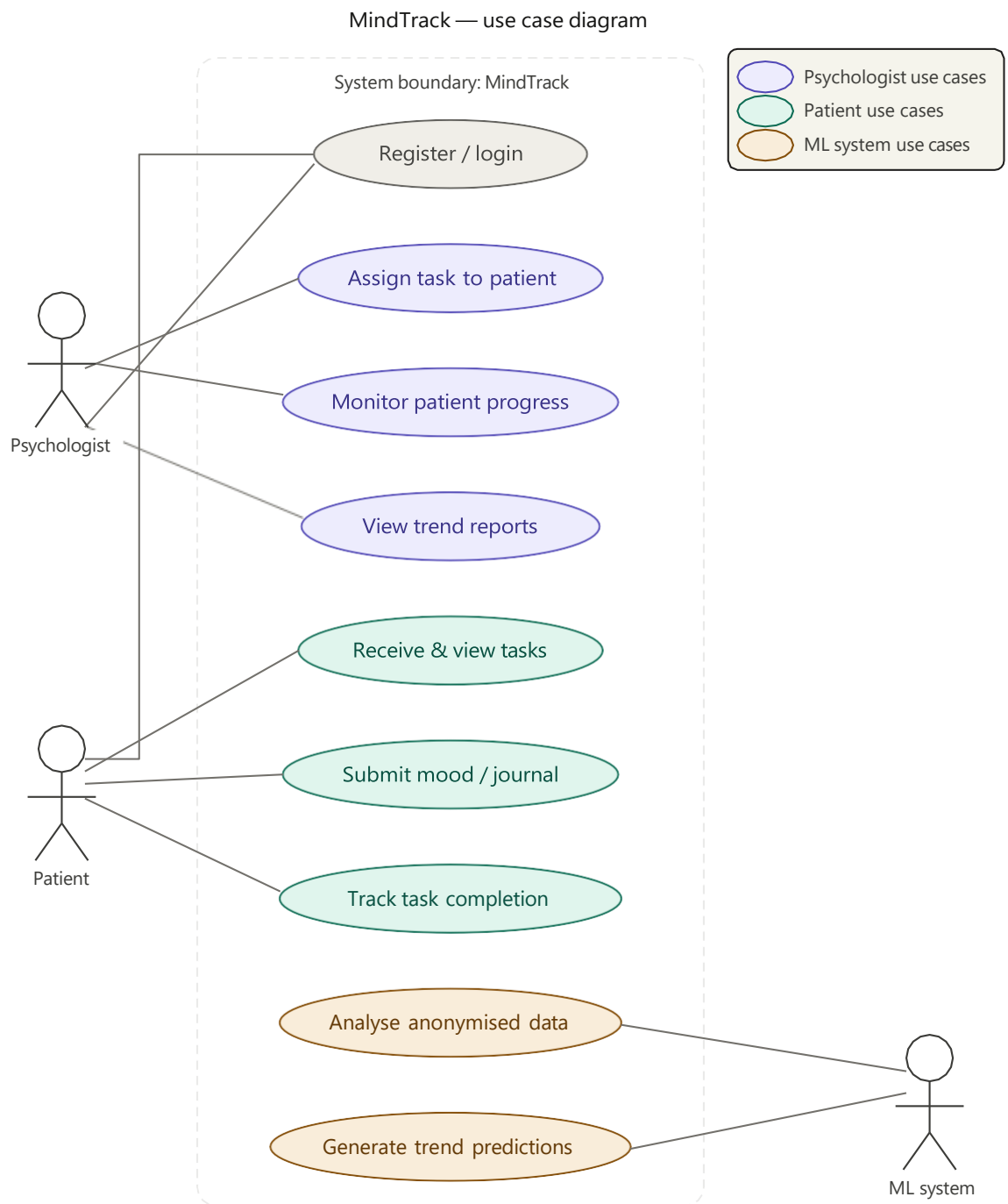


Fig. 6.2: DFD- Level 1



### c. Use Case Diagram



**Fig. 6.3: DFD- Use case diagram**

#### **d. Implementation Details**

The implementation of the MindTrack system was carried out using Android Studio as the primary development environment. The frontend of the application was developed using Java/XML to create interactive interfaces for both psychologists and patients. Separate dashboards were implemented to provide role-based access and functionalities.

Firebase Authentication was integrated for secure login and registration, while Firestore Database was used to store patient records, mood logs, journal entries, assigned tasks, and appointment details. Real-time synchronization was implemented to ensure smooth communication between psychologists and patients.

The application includes modules such as task assignment, mood tracking, journaling, appointment scheduling, and report visualization. Patients can submit weekly emotional data such as mood status (happy, sad, stressed, etc.), while psychologists can monitor patient progress through the dashboard.

For intelligent analysis, machine learning models were developed using Python libraries such as Scikit-learn, Pandas, and NumPy. Unsupervised learning techniques like clustering were applied to analyze anonymized patient data and identify emotional trends and behavioral patterns. NLP-based sentiment analysis was also implemented on journal entries to generate meaningful psychological insights.

Matplotlib was used for generating graphs and visual reports, helping psychologists understand weekly emotional patterns and therapy effectiveness. The system also includes data anonymization and encryption techniques to maintain patient privacy and data security throughout the application.

# RESULT ANALYSIS

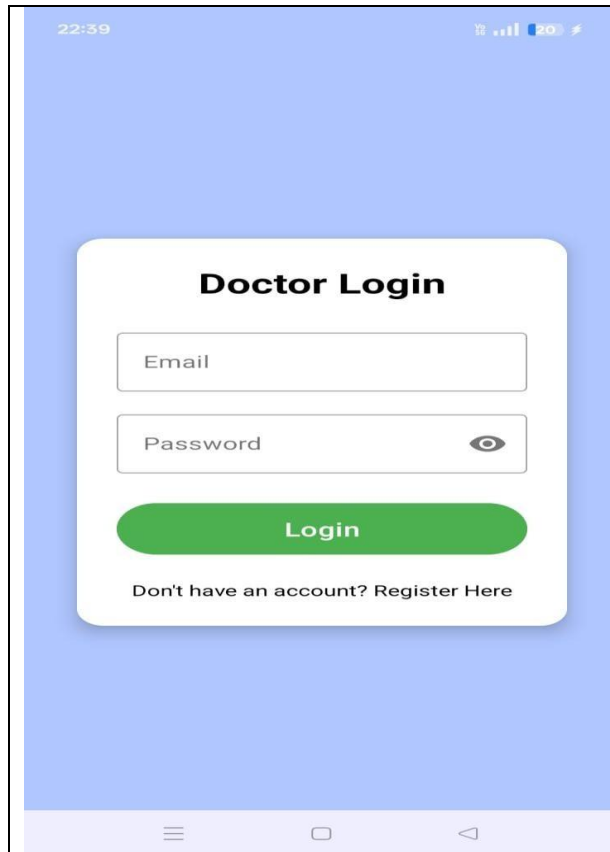


Fig. 7.1

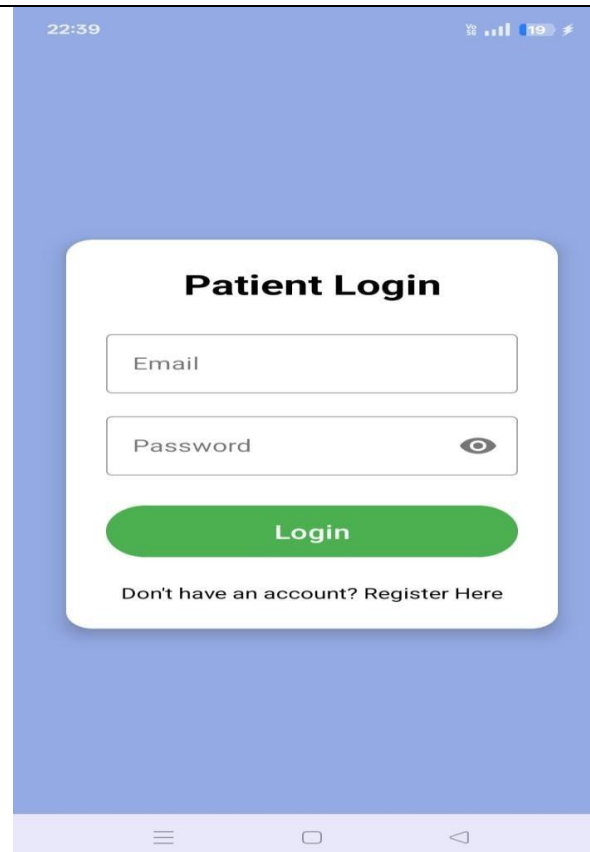


Fig. 7.2



Fig. 7.3

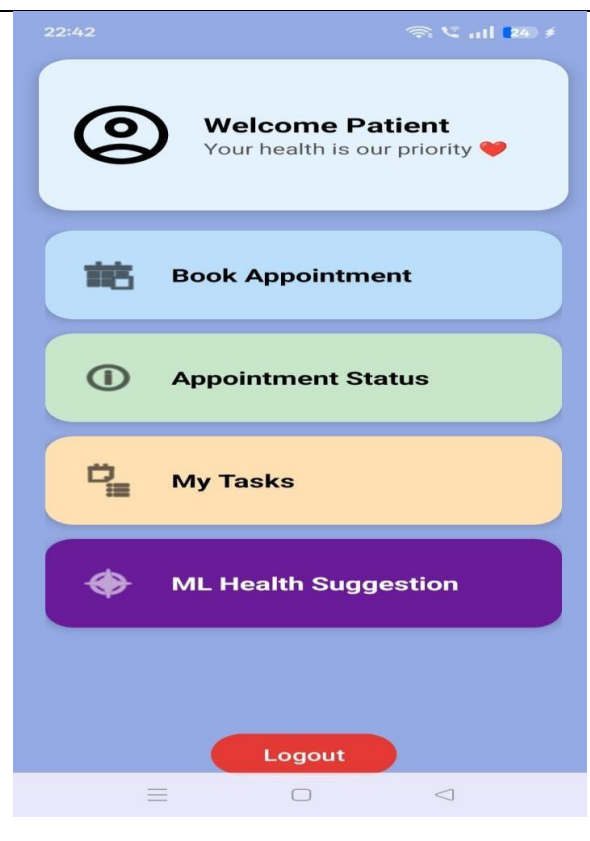


Fig. 7.4

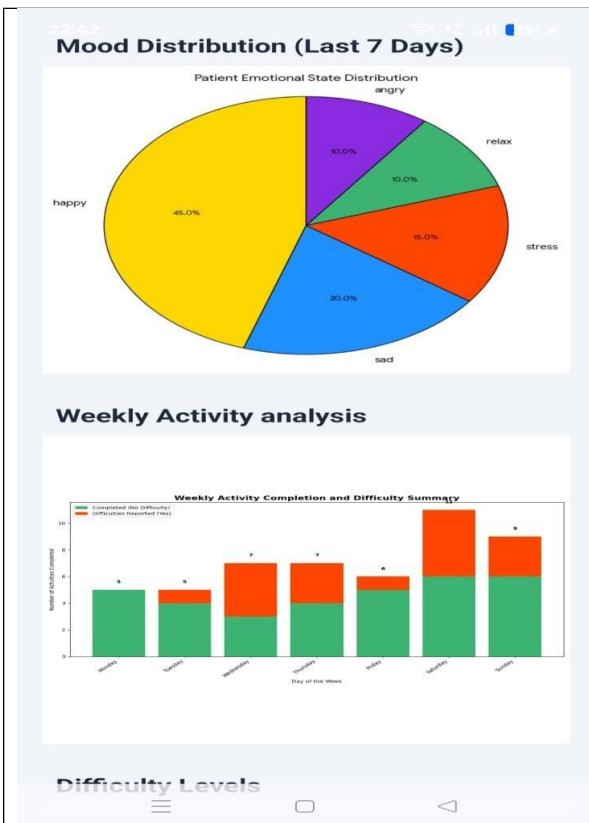


Fig. 7.5



Fig. 7.6

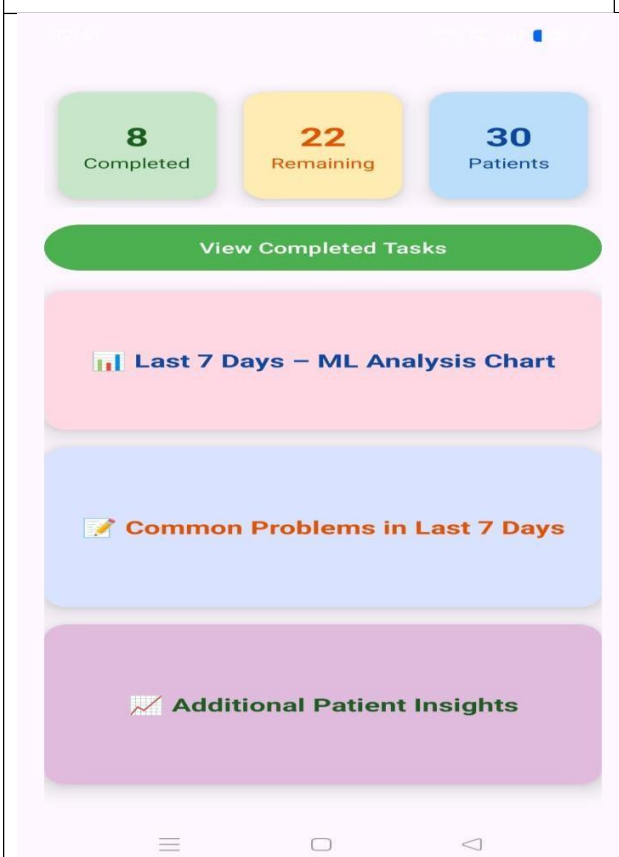


Fig. 7.7

**Create new task**

Assign a task for patient follow-up, therapy reminders or assessments.

Select patient

Task title (e.g., Weekly CBT check-in)

Details (notes, instructions, therapy link...)

Start date Due date

Reference link (video, article, google link)

Additional notes (optional)

SET REMINDER TIME Repeat

CANCEL SAVE TASK

Fig. 7.8

# CONCLUSION & FUTURE SCOPE

## a. Conclusion

The MindTrack project successfully demonstrates the application of modern mobile and machine learning technologies in the field of mental healthcare. The system was designed to improve communication and interaction between psychologists and patients by providing a secure and user-friendly Android platform. Through features such as task assignment, mood tracking, journaling, appointment scheduling, and report generation, the application creates a continuous therapeutic environment beyond traditional therapy sessions.

A major contribution of the project is the integration of machine learning and data analytics for identifying emotional and behavioral trends from anonymized patient data. By analyzing weekly mood patterns, journal entries, and patient activities, the system helps psychologists gain meaningful insights into patient progress and common stress-related patterns. The use of unsupervised learning and sentiment analysis enables the generation of data-driven reports that support better clinical decision-making and personalized care.

The project also emphasizes patient privacy and security through the implementation of secure authentication, encrypted data handling, and anonymization techniques. Firebase services ensure reliable real-time data synchronization, while Python-based ML libraries provide effective analytical capabilities. The graphical visualizations generated by the system help psychologists easily understand emotional trends and therapy effectiveness.

Overall, MindTrack transforms the traditional session-based mental healthcare approach into a proactive, technology-driven system. The project demonstrates how Android development, cloud databases, machine learning, and natural language processing can work together to enhance mental health monitoring and therapeutic support. Future improvements such as wearable device integration, AI chat assistance, predictive mental health alerts, and multilingual support can further expand the system's capabilities and impact in the healthcare domain.

## **b. Future Scope**

- **Integration with Wearable Devices:** The system can be integrated with smartwatches and fitness bands to collect real-time health data such as heart rate, sleep patterns, and stress levels for more accurate mental health analysis.
- **AI-Based Recommendation System:** Advanced machine learning models can be implemented to provide personalized therapy suggestions, mental wellness activities, and task recommendations based on patient behavior and emotional patterns.
- **Predictive Mental Health Alerts:** Future versions of the system can include predictive analytics to detect early signs of stress, anxiety, or depression and automatically alert psychologists for timely intervention.

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## **PUBLICATION DETAILS**

We have published the research paper entitled “MindTrack: Doctor–Patient Mental Health Task Assignment & Prediction” in the International Journal of Scientific Research in Engineering and Management, Volume: 10, Issue: 5 (May 2026), ISSN: 2582-3930, Manuscript ID: IJSREM63433.